Chapter 1 — Introduction

1.1 Mathematical modelling of bacterial resistance to multiple antibiotics and immune system response

1.1.1 Background and Motivation

Bacterial infections have historically been among the leading causes of morbidity and mortality worldwide. The discovery of antibiotics transformed once fatal infections into treatable conditions. However the widespread use of antibiotics has led to the emergence of resistant bacterial strains, which now pose a serious threat to global public health.

Antibiotic resistance occurs when bacteria evolve mechanisms that allow them to survive exposure to drugs designed to kill or inhibit them.

Mathematical modelling offers a powerful approach to studying resistance dynamics. By translating biological interactions into equations, models allow researchers to analyze system behavior, identify critical parameters, and evaluate treatment strategies in a controlled and reproducible way. This project applies such an approach, using a system of ordinary differential equations to model the interaction between sensitive bacteria, resistant bacteria, immune cells, and multiple antibiotics within a single host.

1.1.2 Problem Statement

Despite decades of research, bacterial resistance to antibiotics continues to rise at an alarming rate. Clinical treatment decisions such as which antibiotics to use, at what dose, and for how long are often made without a full quantitative understanding of how resistance develops and how the immune system interacts with both the bacteria and the drugs. Most existing mathematical models either ignore the immune response entirely or treat it as a constant background effect, which fails to capture the dynamic and adaptive nature of host immunity.

Furthermore, the majority of models focus on either the spread of resistance across a population or the pharmacokinetics of a single antibiotic, rarely combining multiple antibiotics with a dynamic immune response in a single within-host framework. This leaves a gap in understanding how these factors interact simultaneously during the course of treatment in an individual patient.

This project addresses that gap by implementing and analyzing the within-host model , which incorporates sensitive and resistant bacterial populations, a dynamic immune cell population, and the concentrations of multiple antibiotics simultaneously. The central question is: under what conditions does antibiotic treatment succeed in controlling infection, and what role does immune system strength play in determining that outcome?

1.1.3 Objectives of the Study

The objectives of this project are as follows:

- To implement the mathematical model numerically using MATLAB.
- To reproduce and verify the analytical equilibrium results through simulation.
- To investigate the effect of immune system strength on treatment outcome.
- To analyze how the fitness cost of resistance influences the persistence of resistant bacteria.
- To evaluate the consequences of early antibiotic termination on bacterial load.
- To compare single antibiotic therapy with combination therapy quantitatively.
- To perform a sensitivity analysis identifying the most critical model parameters.

1.1.4 Scope and Limitations

This study operates within the following scope and is subject to the following limitations:

Scope

- This study focuses exclusively on within-host dynamics — the behavior of bacteria, immune cells, and antibiotics within a single infected individual
- The model considers two antibiotics administered simultaneously as combination therapy
- Analysis covers both mathematical stability and numerical simulation over a 90-day treatment period.

Limitations

- The model does not capture spread of infection between individuals or across hospital populations
- Resistance is assumed to arise solely through spontaneous mutation — horizontal gene transfer and conjugation between bacteria are not modelled
- The immune response is represented by a single population of cells growing logistically in proportion to bacterial load — innate immunity, antibody responses, and cytokine signaling are not included.

1.2 Logistic Population Models with Harvesting

1.2.1 Background and Motivation

The study of population dynamics has been a fundamental concern in both ecology and applied mathematics. Understanding how populations grow, stabilize, and decline is essential for managing natural resources, particularly in fisheries where human harvesting directly influences population behavior.

The earliest formal mathematical treatment of population growth was proposed by Malthus in 1798, who described growth as purely exponential. While this captured the basic idea of biological reproduction, it failed to account for environmental limits such as food supply and space. To address this, Verhulst introduced the logistic growth model in 1838, incorporating the concept of carrying capacity — the maximum population size an environment can sustainably support. The logistic model became the foundation for much of modern population ecology.

In fishery management, the logistic model was further developed to include harvesting — the deliberate removal of individuals from the population. Two major strategies have been widely studied: constant harvesting, where a fixed number of fish are removed per unit time, and periodic harvesting, which reflects the seasonal nature of real fishing activity where grounds are opened and closed at regular intervals.

These mathematical models, expressed as ordinary differential equations, allow researchers and policymakers to analyze long-term population behavior, identify critical thresholds, and design sustainable harvesting strategies that prevent extinction while maximizing yield.

1.2.2 Problem Statement

Fish populations worldwide are under increasing pressure from commercial and subsistence fishing. According to the Food and Agriculture Organization of the United Nations, a significant portion of global fish stocks are either fully exploited or overexploited, meaning they are being harvested at or beyond their biological

limits. When harvesting exceeds the natural capacity of a population to recover, the population enters a state of irreversible decline that can ultimately lead to commercial extinction or even biological extinction of the species.

The central problem addressed in this study is the determination of safe and sustainable harvesting rates for a fish population governed by logistic growth dynamics. Specifically, the study asks: at what harvesting rate does a fish population remain stable, and at what point does harvesting cause the population to collapse? Furthermore, how does the pattern of harvesting — whether constant or periodic — affect the long-term survival and stability of the population?

These questions cannot be answered by intuition or observation alone. A rigorous mathematical framework is needed to model the interaction between natural population growth and human harvesting activity, to identify equilibrium states, to assess their stability, and to determine the critical harvesting threshold known as the Maximum Sustainable Yield (MSY) — the largest harvesting rate that a population can withstand without being driven toward extinction.

1.2.3 Objectives

This study aims to analyze logistic population models under two harvesting strategies using mathematical and numerical methods. Specifically, the objectives are:

1. To derive and analyze the logistic growth model with constant harvesting and determine its equilibrium points using the quadratic formula.
2. To classify the stability of each equilibrium point using linearization and phase portrait analysis.
3. To derive the Maximum Sustainable Yield (MSY) and determine the critical harvesting threshold beyond which population extinction occurs.
4. To formulate and numerically analyze the logistic growth model with periodic harvesting using the Runge-Kutta method (ode45).

5. To compare the long-term behavior of fish populations under constant and periodic harvesting strategies through graphical and numerical simulation.
6. To investigate the sensitivity of the model to changes in key parameters including the growth rate k, carrying capacity N, and harvesting rate a.

1.2.4 Scope and Limitations

This study focuses exclusively on single-species fish population models governed by the logistic differential equation with two forms of harvesting — constant and periodic. The analysis is conducted within the following scope and subject to the following limitations:

Scope:

- The study considers a closed population with no immigration or emigration.
- Two harvesting models are analyzed: constant harvesting and sinusoidal periodic harvesting.
- Equilibrium analysis, stability classification, and numerical simulation are performed for a defined range of parameter values.
- MATLAB is used as the primary computational tool for numerical integration and graphical visualization.

Limitations:

- The model assumes a single isolated species with no predator-prey interactions or competition from other species.
- Environmental factors such as temperature, disease, pollution, and habitat loss are not incorporated into the model.
- The population is treated as a continuous variable, which is a mathematical idealization — in reality, population size is a discrete whole number.

1.3 MATLAB

MATLAB (Matrix Laboratory) is a high-level programming language and numerical computing environment developed by MathWorks. It is widely used in engineering, mathematics, and science for numerical analysis, matrix computation, algorithm development, and data visualization. Unlike general purpose programming languages, MATLAB is designed specifically for mathematical computation, making it particularly well suited for solving and visualizing systems of differential equations.

In this project MATLAB was used for the following purposes:

- **Numerical simulation** — the five equation ODE system was implemented as a MATLAB function and solved using the built-in solver ode45, which uses an adaptive Runge-Kutta method to integrate the system .
- **Parameter management** — all model parameters were organized into a struct object, allowing different simulation cases to be run by changing a single parameter set without modifying the model function
- **Stability verification** — threshold parameters A, B and C were computed numerically and compared against analytical predictions to verify correctness/
- **Visualization** — time series plots, phase portraits, and comparison plots were generated using MATLABs built in plotting functions to illustrate system dynamics across all simulation scenarios
- **Sensitivity analysis** — for loops were used to automate parameter sweeps across multiple values of η, c, and treatment duration, producing the comparison plots.
- **Results export** — numerical results were organized into formatted tables using MATLAB's table function for inclusion in this report